Complaint Management System - MySQL Capstone Project

Project Objective: Design and implement a Complaint Management System database using MySQL to manage users, employees, departments and complaints. Apply normalization, constraints, CRUD, joins, subqueries, views and transactions in a realistic business scenario.

Problem Statement: Organizations receive complaints that must be assigned to the appropriate department and employee for resolution. Build a relational database that stores complaint information and supports operational and reporting queries.

MySQL Workflow

- Requirement Analysis
- Identify entities (Users, Departments, Employees, Categories, Complaints)
- Draw ER Diagram
- Normalize to 3NF
- Create database and tables
- Insert sample data
- Perform CRUD
- Numeric, String & Date functions
- Write joins, subqueries
- Create views
- Perform transactions
- Generate reports

Database Design

Users(user_id PK, full_name, phone UNIQUE, email UNIQUE, address)
Departments(department_id PK, department_name)
Employees(employee_id PK, department_id FK, employee_name, phone, email)
Complaint_Categories (category_id PK, category_name)
Complaints(complaint_id PK, user_id FK, employee_id FK, category_id FK, complaint_date, priority, status, description)

Sample Data

```
-- Departments
INSERT INTO Departments VALUES
(1,'IT'), (2,'HR'), (3,'Finance'), (4,'Network'), (5,'Adminstration');

-- Categories
INSERT INTO Complaint_Categories VALUES
(1,'Software'), (2,'Hardware'), (3,'Network'), (4,'Login'), (5,'Security'), (6,'Other');
```

```
 -- Users
INSERT INTO Users VALUES
(1,'Aarav','900000001','u1@mail.com','Vijayawada'),
(2,'Bhavya','900000002','u2@mail.com','Hyderabad'),
(3,'Charan','900000003','u3@mail.com','Guntur'),
(4,'Divya','900000004','u4@mail.com','Vizag'),
(5,'Esha','900000005','u5@mail.com','Nellore'),
(6,'Farhan','900000006','u6@mail.com','Kurnool'),
(7,'Gopi','900000007','u7@mail.com','Tirupati'),
(8,'Hari','900000008','u8@mail.com','Ongole'),
(9,'Isha','900000009','u9@mail.com','Warangal'),
(10,'John','900000010','u10@mail.com','Chennai');

-- Employees
INSERT INTO Employees VALUES
(101,1,'Ravi','910000001','e1@mail.com'),
(102,1,'Priya','910000002','e2@mail.com'),
(103,2,'Kiran','910000003','e3@mail.com'),
(104,2,'Sneha','910000004','e4@mail.com'),
(105,3,'Arjun','910000005','e5@mail.com'),
(106,4,'Meena','910000006','e6@mail.com'),
(107,5,'Vikas','910000007','e7@mail.com'),
(108,1,'Divya','910000008','e8@mail.com');

-- Complaints
INSERT INTO Complaints VALUES

(1001,1,101,4,'2026-07-01','High','Open','Unable to login to employee portal'),

(1002,1,102,1,'2026-07-02','Medium','Resolved','ERP application crashes frequently'),

(1003,2,103,3,'2026-07-02','High','In Progress', 'Office internet is not working'),

(1004,2,104,2,'2026-07-03','Low','Closed','Desktop not powering on'),

(1005,3,101,5,'2026-07-03','High','Resolved','Unauthorized login attempt detected'),

(1006,3,105,6,'2026-07-04','Medium','Open','Need access to shared folder'),

(1007,4,106,2,'2026-07-04','High','In Progress', 'Printer not detecting system'),

(1008,4,107,6,'2026-07-05','Low','Closed','Request for workstation relocation'),

(1009,5,108,1,'2026-07-05','Medium','Resolved','Payroll software not responding'),

(1010,5,103,3,'2026-07-06','High','Open','VPN connection failed'),
```

(1011,6,104,4,'2026-07-06','Medium','Resolved','Password reset email not received'),

(1012,6,105,5,'2026-07-07','High','Closed','Suspicious email reported'),

(1013,7,106,2,'2026-07-07','Low','Open','Keyboard keys are not working'),

(1014,7,101,1,'2026-07-08','Medium','In Progress', 'MS Excel keeps crashing'),

(1015,8,102,6,'2026-07-08','Low','Resolved','Need new software installation'),

(1016,8,107,3,'2026-07-09','High','Open','LAN cable disconnected'),

(1017,9,108,4,'2026-07-09','Medium','Closed','Unable to login after password change'),

(1018,9,103,5,'2026-07-10','High','Resolved','System blocked due to security policy'),

(1019,10,104,1,'2026-07-10','Low','In Progress', 'Application loading very slowly'),

(1020,10,105,6,'2026-07-11','Medium','Open','General technical assistance required');

30 Business Queries

1. List all complaints.

2. Show complaint with user name.

3. Display all employee names in **uppercase and lowercase.**

4. Display the first **4 characters &** last **5 characters** of every complaint description.

5. Concatenate employee name and department name with and without separator.

6. Replace the word **"login"** with **"authentication"** in complaint descriptions.

7. Find complaints raised in the last **7 days**.

8. Count complaints raised department-wise this month.

9. Display complaint dates in the format: 08-Jul-2026

10. Show complaint with employee and department.

11. Count complaints by status.

12. Count complaints by priority.

13. Department with maximum complaints.

14. Employee handling maximum complaints.

15. Open complaints.

16. Resolved complaints.

17. Complaints raised today.

18. Users with multiple complaints.

19. Categories with more than one complaint.

20. Average complaints per employee.

21. Top 5 latest complaints.

22. Complaints assigned to IT.

23. Users without complaints.

24. Employees without assigned complaints.

25. Show the current status of complaints raised in the last 15 days.

26. Generate a complaint reference by combining the category name and complaint ID (e.g., SOFTWARE-1001).

27. Display the employee name in uppercase along with the number of complaints assigned.

28. High priority open complaints.

29. Create view for open complaints and retrieve data from it.

30. Transaction: assign employee then update status.

31. Create a stored procedure to display all complaints assigned to a particular employee.

- Input: employee_id
- Display complaint_id, priority, status, and description.

32. Create a stored procedure to display all complaints having a particular priority.

- Input: priority
- For example, display all High priority complaints.

33. Create a stored procedure to count the number of complaints for a particular category.

- Input: category_id
- Display the total number of complaints.

34. Create a trigger on the Complaints table.

- Whenever a new complaint is inserted, automatically store the complaint_id, user_id, category_id, and complaint date in a Complaint_Log table.

35. Create a trigger on the Complaints table.

- Whenever the complaint status is updated, store the complaint_id, old status, new status, and update date in a Complaint_Status_History table.